

Garrett Hagen

Software Engineer

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EXPERIENCE

Apple Inc — Apple TV App

iOS Software Engineer · July 2022 – Present

- Own the Apple TV App widget, maintaining it and shipping new features across iOS, tvOS, macOS, and visionOS.
- Built client support for TV App features including condensed and full sports replays and partner-app integration features.
- Designed and own the client instrumentation that measures app launch and page-load performance and reports UI events across the TV App.
- Led the UI and data-modeling layer for an upcoming iOS feature (*under NDA*).
- Delivered Apple TV App functionality for Siri AI, Apple's new Apple Intelligence-powered assistant, collaborating across the AI/ML and Siri teams.

Dexcom Inc — Continuous Glucose Monitoring Systems

iOS Engineer, New Markets Team · Aug 2021 – July 2022

- Owned transmitter integration (G6 and G7) for the hospital CGM iOS app, maintaining and integrating Dexcom's existing Bluetooth LE / CoreBluetooth communication frameworks.
- Built and improved areas of the patient onboarding flow.

QA Engineer, Transmitter Team · Aug 2020 – Aug 2021

- Wrote automated tests exercising real-world usage scenarios between CGM transmitters and mobile apps over Bluetooth LE.
- Built an Appium suite automating typical usage scenarios to reproduce and analyze BLE connection drops, plus tooling to inspect the captured data.

Gener-8 — FDA-Approved CPM Self-Rehabilitation Devices

Software Development & Hardware Design · 2016 – Present

Second-generation system · in progress

- Redesigned the machine hardware around a Raspberry Pi system and rewrote the firmware in Python, adding real-time calibration offsets and improved LED feedback.
- Built an extensive Bluetooth LE feature set into the firmware: headless Wi-Fi configuration, wireless firmware updates, motor-speed control, sensor-state reading, rep counting, and task-progress updates.
- Migrating the app's client logic to defined Firebase Cloud Functions endpoints to centralize business logic and thin the client.
- Moving the codebase from CocoaPods to Swift Package Manager, splitting shared code into reusable Gener-8 packages consumed by both the factory-setup and production apps.
- Wrote Python admin tools and manufacturing software for the platform, spanning system database administration and Raspberry Pi device provisioning.
- Refactored the Gener-8 website using agentic, AI-assisted engineering workflows.
- Built an agentic development harness (Ralph loop) coordinating multiple Claude agents across the BLE peripheral and a paired iOS device — one SSHing into the Gener-8 to tail logs and run commands, another driving and monitoring the iOS app — enabling automated, remote development and debugging.

First-generation system

- Designed the circuitry and embedded software for an on-device angle-calculation system that streams real-time IMU measurements over Bluetooth LE.
- Shipped the Gener-8 iOS app — machine communication, calibration, patient workout-data capture and storage, patient-therapist messaging, and rehab visualization and feedback.
- Owned the full technical stack, from circuit design and manufacturing to app development and App Store deployment.

Carnegie Robotics — Advanced Robotics Sensors & Platforms

Software Co-op · 2018 – 2019

- Created and maintained a sizeable C++ framework that reduced boilerplate for projects using stereo-camera functionality.
- Built a C++ application assessing extrinsic/intrinsic calibration quality for MultiSense cameras using OpenCV, Ceres-Solver, Boost, and GTest.
- Developed an interactive Qt/OpenCV C++ GUI for concurrent grayscale, color, and disparity streaming across all MultiSense camera models.

SKILLS

Languages: Swift, Python, Objective-C, TypeScript, Java, C++

iOS / Frameworks: CoreBluetooth (BLE), SwiftUI, UIKit, WidgetKit, AppIntents, Firebase, Xcode

Tools: Git, CMake, Appium, JUnit, AutoDesk Inventor, Claude Code (agentic dev)

Platforms: iOS, tvOS, macOS, visionOS, Linux

EDUCATION

University of Pittsburgh

BS, Computer Engineering · 2016–2020

- Major GPA 3.82 · Cumulative 3.55
- Engineering Dean's Honor List (2017–2020)

PROJECTS + OPEN SOURCE

wpa-pyfi 2021

- Open-source Python package to manage Raspberry Pi Wi-Fi via CLI or library. Published on PyPI.

OmniBot 2020

- Omnidirectional capstone robot; built iOS app + BLE protocol for gesture, joystick, and autopilot control.

Biometric Lock 2019

- Swift iOS app over BLE to an Arduino-driven fingerprint scanner + solenoid lock; designed circuitry and custom enclosure.

Industrial Arduino 2017

- Custom PCB shield in Altium for a waterproof Arduino enclosure.

VOLUNTEERING

HCEF Instructor 2019–2020

- Taught basic computer science to middle-school students experiencing homelessness.